GGG-S-278B
March 14, 1994
SUPERSEDING
GGG-S-00278A (GSA-FSS)
June 16, 1964

FEDERAL SPECIFICATION

SHEARS AND SCISSORS

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers shears and scissors generally used for various cutting, trimming, and clipping tasks (see 6.1).

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classifications.

1.2.1 Types, classes, designs, and styles.

Type I - Shears

Class 1	-	Straight trimmers
Design 1	-	Paper or stationers'
Design 2	-	Household or industrial
Class 2	-	Bent trimmers
Class 3	-	Pinking
Class 4	-	Tailors' (Sailmakers')

Type II - Scissors

Class 2	-	Ladies' (sewing)
Class 3	-	Pocket
Class 6	-	Electricians'

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the currentversion before use.

1.2.2 Sizes. The shears and scissors shall be of the sizes specified in tables I through VIII.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40	-	Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28	-	Screw-Threads Standards for Federal Services
FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-H-15424	-	Hand Tools; Packaging of

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

ASTM A 754	-	Coating Thickness by X-Ray Fluorescence
ASTM B 487	-	Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section
ASTM B 499	-	Measurement of Coating Thickness by the Magnetic Method: Nonmagnetic Coatings of Magnetic Basis Metals
ASTM B 504	-	Measurement of Thickness of Metallic Coatings by the Coulometric Method
ASTM B 530	-	Measurement of Coating Thickness by the Magnetic Method: Electrodeposited Nickel Coatings on Magnetic and Nonmagnetic Substrates
ASTM B 568	-	Measurement of Coating Thickness by X-Ray Spectrometry
ASTM B 748	-	Measurement of Thickness Coatings by Measurement of Cross Section with a Scanning Electron Microscope
ASTM E 18	-	Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Industrial Fastener Institute (IFI):

IFI-124	-	Test Procedure for the Locking Ability Performance of Non-Metallic Locking Element Type Prevailing-Torque Lock Screws

(Applications for copies should be addressed to Industrial Fastener Institute, 1505 East Ohio Building, 1717 East 9th Street, Cleveland, OH 44114.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of shears or scissors otherwise conforming to this specification.

3.2 Marking. Each pair of shears or scissors shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

3.3 Materials. Materials used in the construction of shears and scissors shall be of the highest commercial quality, fully suited for each particular application and as specified herein. Materials shall be free from any defects and imperfections which might affect the serviceability of the finished product.

3.4 Threaded parts. Threaded parts shall be in conformance with the applicable requirements of FED-STD-H28. At the option of the manufacturer, torque locking screws which meet or exceed IFI-124 specifications shall be acceptable.

3.5 Coatings. Unless otherwise specified (see 6.2), shears and scissors shall be coated with one of the following coatings at the option of the manufacturer:

(a) Japanned handles with chrome-plated blades
(b) Fully nickel-plated blades and handles
(c) Fully chrome-plated over nickel-plated blades and handles
(d) Japanned handles with nickel-plated blades
(e) Japanned handles with chrome-plated over nickel-plated blades
(f) Comfort grip handles made of a polymer of rubber or plastic with chrome-plated over nickel-plated blades or nickel-plated blades.

3.5.1 Chromium plate. The coating shall have an electrodeposited coating of chromium not less than 0.000003 inch thick when tested as specified in 4.6.3.

3.5.2 Nickel plate. The coating shall have an electrodeposited coating of nickel not less than 0.00008 inch thick when tested as specified in 4.6.3.

3.5.3 Chromium over nickel plate. The coating shall have electrodeposited metals consisting of nickel followed by chromium with the minimum thickness being 0.00008 inch for nickel and 0.000003 inch for chromium when tested as specified in 4.6.3.

3.5.4 Japanning. The coating shall be glossy lacquer, varnish or enamel paint. The coating shall be adherent when tested as specified in 4.4.4.

3.6 Blades.

3.6.1 Blade hardness. Unless otherwise specified herein (see 3.9.3), the blade hardness of all shears and scissors shall show a Rockwell C hardness (HRC) of not less than 52 nor more than 61 when tested as specified in 4.6.2 and 4.6.2.1. Matching points on the blades of a pair of shears and scissors be within for points (hardness).

3.6.2 Joining blades of type I shears. The blades of type I shears shall be joined by a steel bolt or screw or steel bolt or screw and a steel bushing, except for type I, class 2 bent trimmers (10 inch and 12 inch sizes), class 3 pinking shears, and class 4 tailor's shears where a steel bolt with nut shall be used.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the currentversion before use.

3.6.3 Joining blades of type II scissors. The blades of type II scissors shall be joined by a steel bolt or screw except type II, class 3 pocket scissors which shall be joined by a square-shouldered rivet locked in one blade or other suitable means.

3.6.4 Blade adjustments. The shears and scissors shall be adjusted by the manufacturer for proper contact along the entire length of the blade cutting edges. The pressure between the blades shall not be such as to cause binding, blemishes, or other defects observed when cutting tests are conducted as specified in 4.7.1. Type I, class 4 tailors' shears shall be provided with a means for adjusting the pressure between the blades.

3.6.5 Finish. Scissors and shears shall be smoothly and fully finished with no rough or irregular edges and surfaces. All surfaces shall be plated or coated and plated as specified herein except the cutting edges and the insides of the blades may be unplated. Excess flash or welds, or any other defects which detract from the appearance or which may impair the serviceability are not acceptable. The micro inch finish of the plated surfaces shall be 15 maximum, arithmetic average. A brush finish is not acceptable.

3.7 Length of cut. The minimum length of cut shall be as specified in tables I through VIII and shall be defined as the cutting edge of blades measured from the outer end of the blade (tip) to the point at which the blades cross one another in the fully opened position.

3.8 Handles. Unless otherwise specified, handles shall be designed for a right-hand operator.

3.8.1 Type I shears. All type I shears shall be furnished with the large or handgrip all-metal handles (fingers and thumb bows).

3.8.2 Type II scissors. All type II scissors shall be furnished with either double-ring or symmetrical all-metal bow handles.

3.9 Type I shears.

3.9.1 Type I, class 1, straight trimmers, design 1 and design 2. The Type I, design 1 shears shall have both blades pointed, or when specified (see 6.2), one blade pointed and one beveled. Design 2 shears shall have the blade with the thumb ring pointed and the other blade beveled.

3.9.1.1 Design 1, paper or stationers'. The design 1 shears shall be of steel, forged, and of either inlaid or solid steel construction. When inlaid construction is used, the inlaid piece for each blade shall be high carbon steel having .85 percent carbon (minimum) and be securely joined by welding to the blade and handle forming a single integral unit. When solid steel construction is used, blades and handles shall be of one of the following: (1) high carbon steel having a designation of AISI 1060 or greater, or (2) alloy steel having a designation of AISI 4150 or greater. Design 1 shears shall conform to table I and be similar to figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

TABLE I. Type I, class 1, design 1 straight trimmers, paper or stationers' shears

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
9	4-5/8	3
10	5-1/2	4-1/4
12	6-3/4	7
14	8-1/2	9

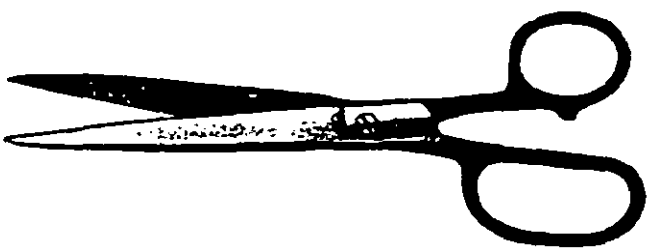


Figure 1. Type I, class 1, design 1 straight trimmer, paper or stationers' shears.

3.9.1.2 Type I, design 2, household or industrial. The type I, design 2, household or industrial shears shall be as specified in 3.9.1.1 except design 2 shears shall conform to table II and be similar to figure 2.

TABLE II. Type I, class 1, design 2, straight trimmers, household or industrial shears

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
6	2-1/2	1-1/2
7	3	1-3/4
8	3-5/8	2-7/8
9	4	5

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the currentversion before use.

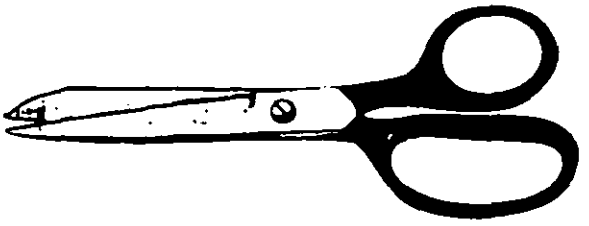


Figure 2. Type I, class 1, design 2, straight trimmer, household or industrial shears.

3.9.2 Type I, class 2, bent trimmers shears. The type I, class 2, bent trimmers shears shall be designed with bent handles and shall have either both blades pointed, or when specified (see 6.2), the blade with the thumb ring shall be pointed and the other blade beveled.

3.9.2.1 Type I, shears. The type I shears shall be as specified in 3.9.1.1 and shall conform to table III and be similar to figure 3.

TABLE III. Type I, class 2, bent trimmers shears

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
7	3-1/8	1-1/4
8	3-1/2	3
9	4	5-1/4
1/ 10	4-1/2	6-1/4
12	5-7/8	10-3/4

1/ For left-hand operation when specified (see 6.2).

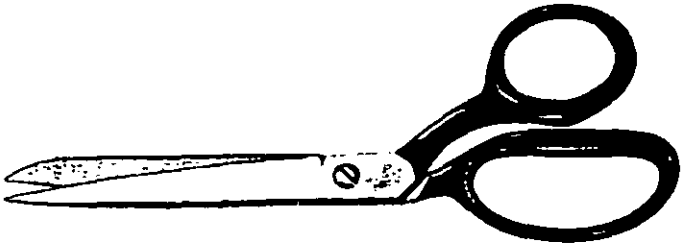


Figure 3. Type I, class 2, bent trimmer shears.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

3.9.3 Type I, class 3, pinking shears. The type I, class 3, pinking shears shall be designed to leave saw-toothed edges (to prevent unraveling of cloth). The blades shall show an HRC or not less than 35. Both blades shall be blunt.

3.9.3.1 Type I, class 3, shears. The type I, class 3, shears shall be of forged solid steel having a minimum carbon content of 0.35 percent. The shears shall conform to table IV and be similar to figure 4.

TABLE IV. Type I, class 3, pinking shears

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
7-1/2	2-5/8	4-7/8
9	3-1/2	5-3/4
11	3-7/8	8-3/4

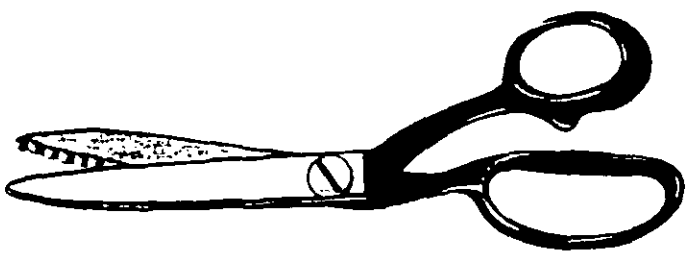


Figure 4. Type I, class 3, pinking shears.

3.9.4 Type I, class 4, tailors' (sailmakers') shears. The type I, class 4, tailors' (sailmakers') shears shall be designed with the tip end of the blade having the thumb ring pointed and the tip of the other blade either beveled, blunted, or pointed. The finger bow shall provide a protrusion suitable for use as a table rest, and the thumb ring shall provide a handle stop. In addition, the thumb ring and adjacent handle area shall be shaped so that when the thumb is inserted as far into the ring as possible, there shall be a smooth contoured ledge provided to support at least 3/4 of the entire length of the thumb. The class 4 shears shall meet the material requirements of 3.9.1.1, shall conform to table V, and be similar to figure 5.

TABLE V. Type I, class 4, tailors' (sailmakers') shears

Overall nominal length (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
12	6	14

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

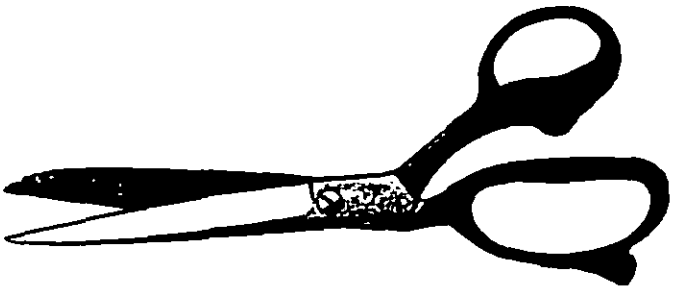


Figure 5. Type I, class 4, tailors' (sailmakers') shears.

3.10 Type II scissors. All type II scissors shall be forged from steel with a minimum carbon content of .55 percent.

3.10.1 Type II, class 2, ladies (sewing) scissors. The type II, class 2, ladies "sewing" or "trimming" scissors shall be designed with one blade pointed and the other blade blunted.

3.10.1.1 Type II, scissors. The type II scissors shall be as specified in 3.10. The scissors shall conform to table VI and be similar to figure 6.

TABLE VI. Type II, class 2, ladies (sewing) scissors

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
4	1-5/8	1/2
5	2-1/4	7/8
6	2-5/8	1-3/8

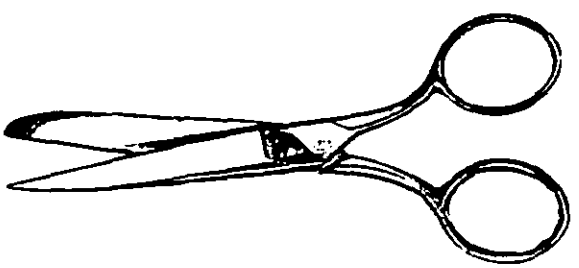


Figure 6. Type II, class 2, ladies' (sewing) scissors.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

3.10.2 Type II, class 3, pocket scissors. The type II, class 3, pocket scissors shall be designed with the points of both blades rounded.

3.10.2.1 Type II, scissors. The type II, class 3 scissors shall be as specified in 3.10. The scissors shall conform to table VII and be similar to figure 7.

TABLE VII. Type II, class 3, pocket scissors

Overall length Nominal (inches)	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
4	1-1/2	3/4
4-1/2	1-3/4	1
5	1-7/8	1-3/8
6	2-5/8	1-7/8

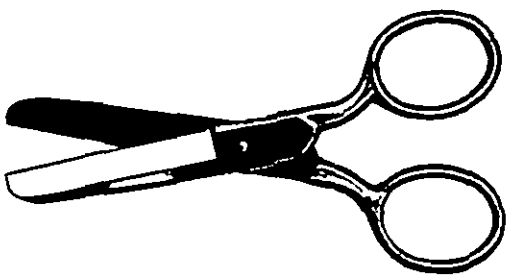


Figure 7. Type II, class 3 pocket scissors.

3.10.3 Type II, class 6, electricians' scissors. The type II, class 6, electricians' scissors shall be designed with both blades pointed and shall have dulled, serrated scraping edges on the backs of the blades. The class 6 scissors shall conform to table VIII and be similar to figures 8A and 8B.

TABLE VIII. Type II, class 6, electricians' scissors

Overall length (inches) Nominal	Actual	Minimum length of cut (inches)	Minimum weight per dozen (pounds)
5	5-1/8 (+1/8 -0)	1-3/4	1-1/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

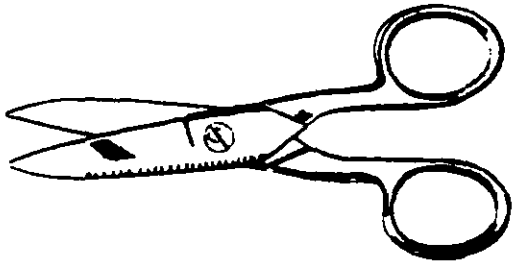


Figure 8A. Type II, class 6, electricians' scissors.

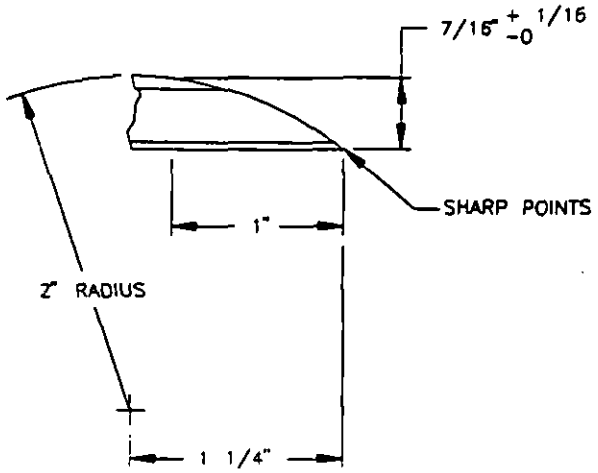


Figure 8B. Type II, class 6, electricians' scissors, point configuration.

3.11 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.12 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 End product inspection. Inspection of the shears and scissors shall be performed as specified in the following paragraphs. The sample unit shall be one completely fabricated shear or scissor. The lot size shall be expressed in units of shears or scissors of one type, class, style, and size offered for inspection at one time for the purpose of determining the sample size in accordance with MIL-STD-105.

4.2.1.1 Visual examination. Each sample unit shall be examined for defects set forth in table IX. The lot size shall be expressed in terms of shears or scissors. The sample unit shall be one pair of shears or scissors. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

TABLE IX. Examination of defects

Examine	Defect	Classification Major	Minor
Coating	Missing or not as specified	101	
	Evidence of rust	102	
Design	Any characteristic not in accordance with specified requirements	103	
Construction	Part missing		104

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:38Z
Check the source to verify that this is the current version before use.

TABLE IX. Examination of defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship	Sharp burr or edges that may be injurious to personnel		105
	Not joined as specified	106	
	Finish not as specified	107	
	Loose or not in proper alignment	108	
	Any adjustable part that cannot be adjusted or is not properly adjusted to perform the function intended	109	
	Functional part that requires abnormal force to operate		110
	Cracked, broken, malformed, bent, bowed, or otherwise damaged	111	
	Ragged or damaged cutting edges	112	
Threads	Stripped, broken, or damaged	113	
Marking identification	Missing, incomplete, not legible, not permanent, or not as specified		201

4.2.2.2 Dimensional and weight examination. Each sample unit shall be examined to determine compliance with the dimensional and weight requirements specified herein. Any dimension or weight that is not within the specified requirements shall be classified as a defect. The lot size shall be expressed in terms of shears or scissors. The sample unit shall be one pair of shears or scissors. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0. The sample unit for the weight examination shall be one dozen, and the sample size shall be S-3 with no AQL.

4.3 Testing. Each sample unit shall be tested for cutting performance to the applicable tests specified under 4.6 to determine compliance with the applicable requirements specified in 4.7. In addition, each sample unit shall be tested to the requirements specified in 4.6 for hardness and plating thickness. The inspection level for the cutting and hardness tests shall be S-3 with an AQL of 1.0 percent expressed in defects per hundred units. The inspection level for the plating thickness test shall be S-1 with an AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table X. The lot size shall be the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Inspection of preparation for delivery for type I, class 4, tailors' (sailmakers') shears shall be in accordance with MIL-H-15424.

TABLE X. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior) method	Omitted; incorrect; illegible; improper size, location, sequence, or of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

4.5 Carbon content. The supplier shall furnish certification that the chemical composition of each lot or batch of steel used in the end item meets the applicable carbon requirements as specified in 3.9.1.1, 3.9.3.1, and 3.10.

4.6 Lot acceptance tests.

4.6.1 Cutting tests. Each of the sample shears or scissors taken in accordance with 4.3 shall be tested as specified in 4.7.1 to verify compliance with the requirements of this specification. Any sample shears or scissors which do not meet the requirements for the cutting tests shall be rejected, and if the number of nonconforming shears or scissors in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6.2 Hardness tests. The HRC tests shall be conducted in accordance with ASTM E 18.

4.6.2.1 Blade hardness. The blades of each sample shear shall be tested to determine compliance with the hardness requirements specified in 3.6.1 and 3.9.3. Readings shall be taken within 1/8 inch from the cutting edge.

4.6.3 Plating thickness test. Each sample shear or scissor shall be tested to determine compliance with the plating thickness requirement specified in 3.5.1, 3.5.2, and 3.5.3. The plating thickness shall be measured on the outside of the blade from the approximate middle of the blade. One measurement

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the current version before use.

of thickness of each electroplated coating shall be conducted in accordance with ASTM A 754, ASTM B 487, ASTM B 499, ASTM B 504, ASTM B 530, ASTM B 568, and ASTM B 748, as applicable. Visible contact marks resulting from electroplating operation s shall be confined to the interior of the shear or scissor handle. Surfaces of the electroplated tool shall be adherent, smooth, continuous, and free of visible defects such as blisters, pits, roughness, cracks, and uncoated areas. Surfaces shall not be stained or discolored and shall have a bright appearance.

4.6.4 Coating adhesion test. The adhesion of lacquer, varnish, or enamel paint shall be tested by applying and removing pressure sensitive tape over cuts made in the coating. The cuts shall be located on a flat area of the tool of sufficient size for the test. When multiple areas, adequate for the test are available on the tool, the testing location shall be varied from one test sequence to the next. The test procedure shall be as follows:

a. Three lines shall be cut completely through the coating. The three lines shall be 1/2 inch ± 1/16 inch long and 1/16 inch to 3/32 inch apart.

b. Another four lines shall be cut completely through the coating overlapping and at 90° ± 10 ° to the first set of lines. These four lines shall be 1/4 inch ± 1/16 inch long and 1/16 inch to 3/32 inch apart.

c. The cut pattern shall be six 1/16 inch to 3/32 inch approximate squares/rectangles.

d. A length of semitransparent tape having an adhesion strength of 40 ounces per inch of width, ± 2.5 ounces per inch of width, shall be firmly pressed over the cut pattern. All air pockets under the tape shall be eliminated and the tape shall be firmly adherent to the coating.

e. After 90 seconds, ± 30 seconds, the tape shall be stripped from the coated surface by rapidly pulling the tape back upon itself at 180 ° ± 30 °.

f. The removal of one or more squares/rectangles of coating shall be cause for rejection.

4.7 Test procedures.

4.7.1 Cutting tests. The shears and scissors shall be tested as follows:

4.7.1.1 Hand cutting test. Each sample shall be submitted to hand-cutting tests using the following materials:

(a) Silk cloth, plain weave.
(b) Tissue paper, basis weight 11.6 pounds ± 5 percent per 500 sheets, size 25 by 40 inches.
(c) Cotton sheeting, plain weave, 3.8 ounces per square yard minimum, and 60 x 60 threads per inch ± 2 threads per inch.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:38Z
Check the source to verify that this is the currentversion before use.

4.7.1.2 Finish. The material in the following tests shall be cut cleanly, without any drag or pull, and shall fall freely away from the cutting edges, including the points.

4.7.1.3 Cuts. Tests as specified in 4.7.1.4 shall consist of three cuts in the following order:

(1) Using only the extreme end of the cutting blades, beginning one inch from the point of the blades.
(2) The full cutting edges (see 3.7).

4.7.1.4 Thickness. The cutting tests shall be made as follows and in the order given:

(1) Cut one thickness of silk cloth.
(2) Cut one thickness of tissue paper.
(3) Cut one thickness of cotton sheeting.
(4) Cut multiple thicknesses of cotton sheeting, for 3/4 of length of the cutting blade measured from the point, using the number of thicknesses prescribed in table XI (see 3.7).
(5) Cut one thickness of cotton sheeting.
(6) Cut one thickness of tissue paper.
(7) Cut one thickness of silk cloth.

TABLE XI. Thickness of cuts

	Overall length nominal (inches)				
	Up to 6	6 to 8	9	10	Over 10
Type I, class 1, design 1	---	---	3	3	4
Type I, class 1, design 2, class 2	---	6	8	12	14
Type II, all scissors except class 6	4	---	---	---	---

4.7.1.5 Type I, class 3, pinking shears. The cutting test for the type I, class 3, pinking shears shall consist of cutting one thickness of silk and three thicknesses of cotton sheeting for the full length of the toothed edge after engagement of the first tooth.

4.7.1.6 Type I, class 4, tailors' (sailmakers') shears. The cutting test for the type I, class 4, tailors' (sailmakers') shears shall consist of cutting 25 thicknesses of cotton sheeting for 3/4 of the length of the cutting blade measured from the point.

4.7.1.7 Tests for type II, class 6, electricians' scissors.

4.7.1.7.1 Cotton sheeting cutting test. Each sample scissors shall be submitted to hand cutting tests using cotton sheeting, plain weave, 56 x 60 threads per inch and 3.8 ounces per square yard. This test material shall be cut cleanly, without drag or pull, and shall fall freely away from the cutting edges, including the points.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

4.7.1.7.2 Aluminum oxide cutting test. Each sample scissors shall be required to make a total of 150 cuts in 150 grit aluminum oxide single-faced abrasive cloth. After every 25 cuts, the scissors shall be tested for cutting one thickness of 3.8 ounce cotton sheeting (see 4.7.1.7.1). During the next 25 cuts, the opposite blade shall face the abrasive side of the aluminum oxide cloth. Failure to cut the 3.8 ounce cotton cloth cleanly for at least 3/4 of the length of the blade, measured from the tips, shall be cause for rejection. The tests requiring 25 cuts in the aluminum oxide described above shall be performed by hand; however, after every 25 cuts (by hand) the tests for cutting one thickness of 3.8 ounce cotton sheeting (see 4.7.1.7.1) shall be performed on a testing machine like or similar to that shown in figure 13 and described in paragraph 4.7.1.7.2.1.

4.7.1.7.2.1 Scissors testing machine. The testing machine shall be designed to clamp the screw that joins the scissors between two pins and retain one blade in the horizontal position by holding the thumb loop on a fixed mandrel. The finger loop at the other blade shall be positioned on a movable mandrel which shall be actuated by a variable speed motor through a mechanical linkage system. The speed of the motor shall produce 75 $\pm$ 10 percent cuts per minute. The mechanical linkage system shall force the movable blade fully open (not to exceed 90° included angle) and then fully close the scissors (so that points meet or overlap slightly) in each cutting cycle (see figure 9). A string or light cable shall be attached, as indicated in the figure to the end of the finger loop of the movable blade, and a 100 gram weight shall be attached to the other end. The cable and weight shall be positioned over a pulley located three feet from, and on the center line, of the screw that joins the scissors so that the 100 gram weight acts in the direction to pull the two blades together assimilating side thrust.

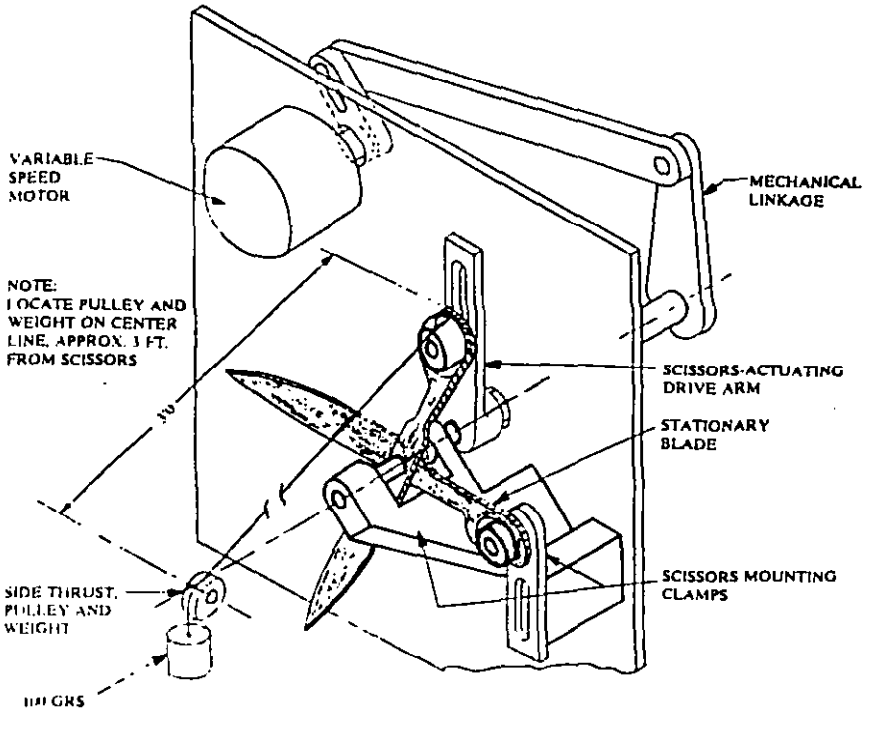


Figure 9. Scissors testing machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

4.7.1.7.3 Wire cutting test. Using sample scissors not already tested in 4.7.1.7.2, take 20 cuts through wire consisting of 10 strands of copper #36 (0.0050) AWG or heavier. The scissors under test shall cleanly cut the wire without any assistance such as bending, pulling, or twisting of the wire by the operator. There shall be no visible deformation or damage to the cutting edges of the scissors.

4.7.1.7.4 Screw torque test. Each sample electricians' scissors shall be tested for "back off" torque put on the joining screw. A torque screwdriver or a torque wrench with a screwdriver attachment shall be used and the joining screw shall withstand a torque of 10 inch-pounds without backing off.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, the interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military activities. In addition to any special markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use.

6.1.1 Type I, shears or scissors. The type I, style A, shears or scissors are recommended when cutting requirements are of such a specialized nature than only the best quality all steel (forged, solid, or inlaid) shears or scissors will perform satisfactorily.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and design of shears and scissors required (see 1.2.1).
c. Size of item required (see 1.2.2).
d. One blade pointed and one beveled (see 3.9.1).
e. The blade with the thumb ring pointed and the other blade beveled (see 3.9.2).
f. For left-hand operation (see note on table III).
g. Level of preservation, packaging, and packing required (see 5.1).
h. Marking, if other than specified (see 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

6.3 Superseding data. The types, classes, designs, and styles in this specification, compared with those covered in Federal Specification GGG-S-278A, are as follows:

TABLE XII. Superseding data

GGG-S-278B	GGG-S-278A
Type I, shears	Type I, shears
Class 1	Class 1
Design 1	
Deleted	Style A
Deleted	Not covered
Design 2	
Deleted	Style B
Deleted	Not covered
Class 2	Class 2
Deleted	Style A
	Style B
Deleted	Not covered
Class 3	Class 3
Deleted	Not covered
Deleted	
Class 4	Class 4
	Style A
	Style B
Type II, scissors	Type II, scissors
Deleted	Not covered
Deleted	
Class 2	Class 2
Deleted	
Deleted	Not covered
Class 3	Class 3
Deleted	Style A
Deleted	Style B
Class 6	Class 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.

MILITARY COORDINATING ACTIVITY:

Army - GL

Custodians:

Army - GL
Air Force - 99

Review Activity:

Air Force - 82

User Activity:

Navy - MC

PREPARING ACTIVITY:

GSA-FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:38Z
Check the source to verify that this is the current version before use.